

WEEKLY JOURNAL CLUB · YOU RECORD · THE PAPER DOES THE TALKING



AI in Medicine Paper Club

Episode 1 — When the LLM reaches the clinic

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THIS WEEK'S PAPER

A generative LLM in live primary care

Agweyu et al. · *Nature Medicine* · 26 June 2026

GPT-4o inside 16 Kenyan clinics. ~10k patients. A hard endpoint.



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doi:10.1038/s41591-026-04503-6





WHY THIS PAPER

Exams are not clinic



Vignettes look easy

LLMs crush tests. Clinic is messier.



60% of amenable deaths

In LMICs, after people already reached care.



Clinical officers

3-year diploma. Often no senior next door.

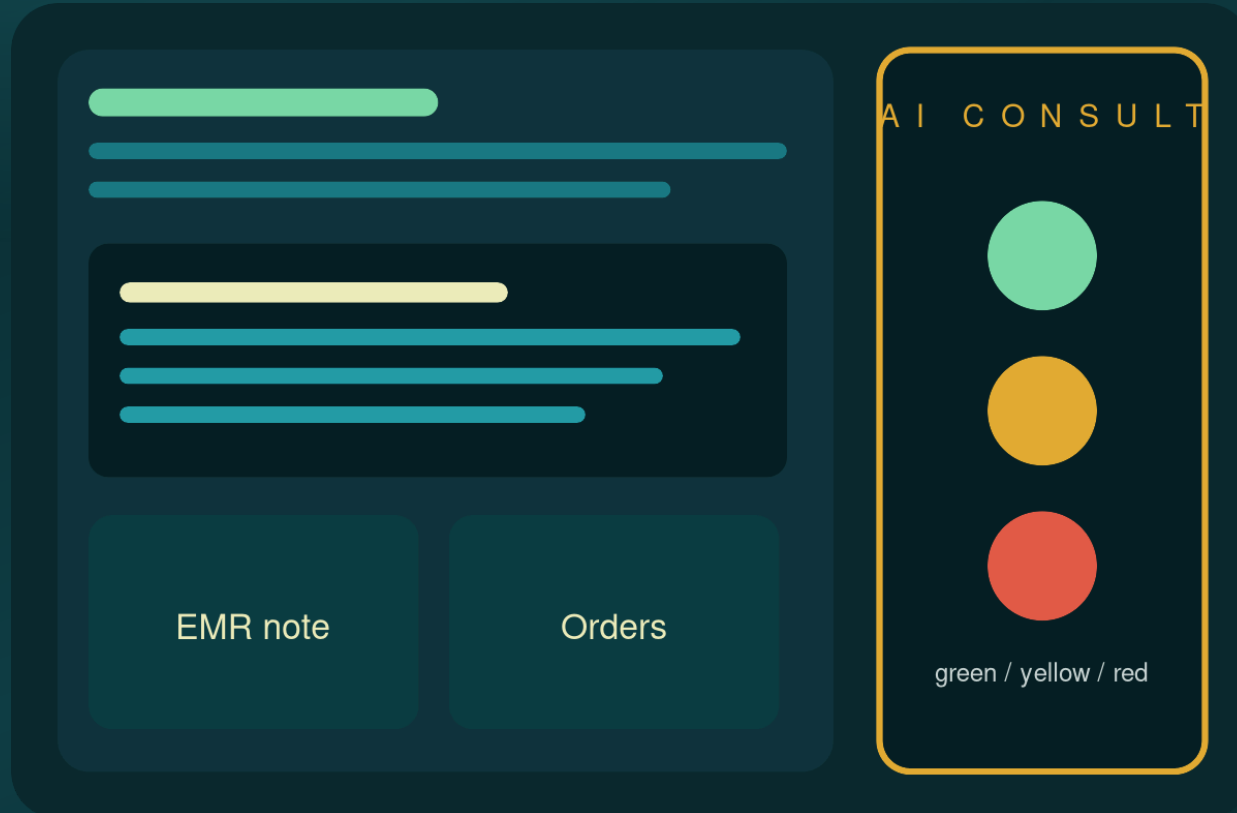


0.3 physicians / 1,000

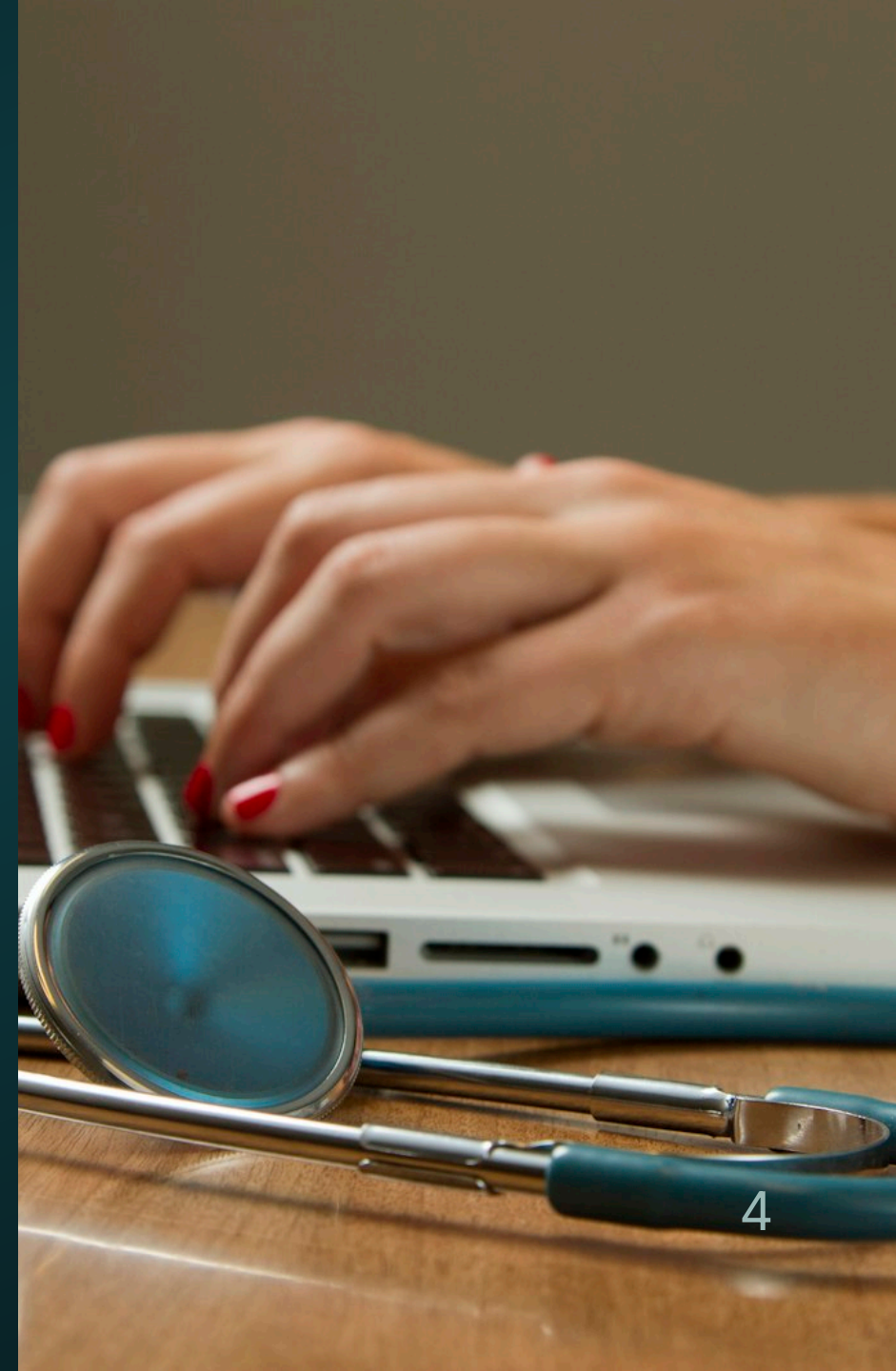
Sub-Saharan Africa vs OECD 3.9.

THE INTERVENTION

AI Consult 2.0 in the EMR

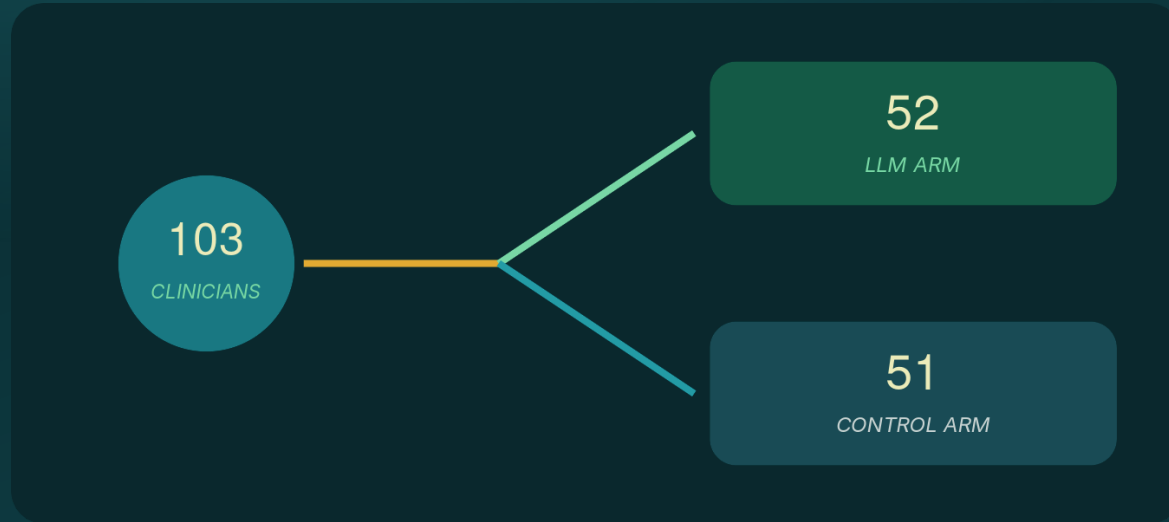


GPT-4o · May 2025 · **US\$0.04** / patient · accept, edit, or ignore



DESIGN

Pragmatic cluster RCT



16 clinics

Nairobi & Kiambu



9,347

encounters analysed



14 days

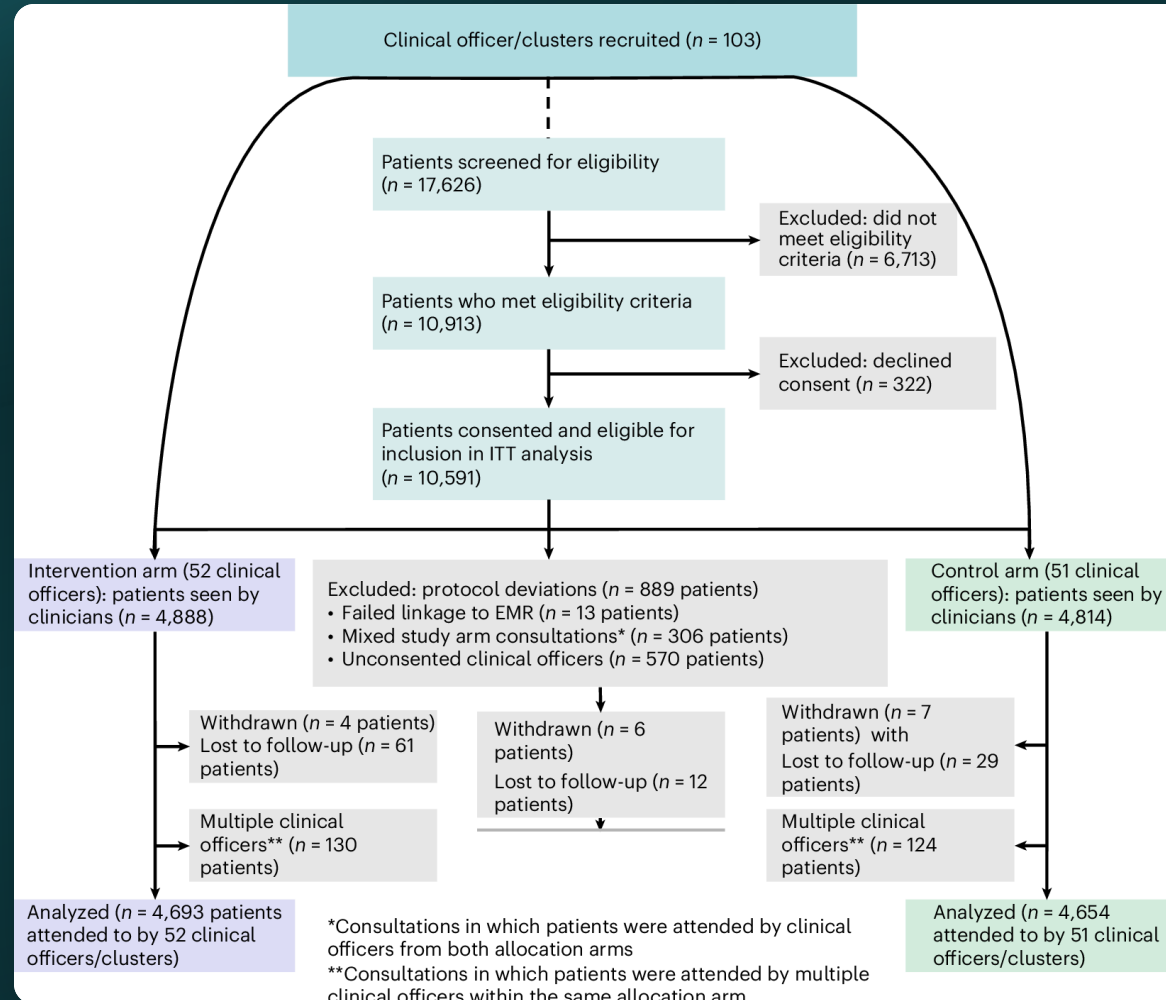
treatment-failure window



Apr-Jul 2025

powered for a 50% drop

Fig. 1 — CONSORT: clusters, then patients

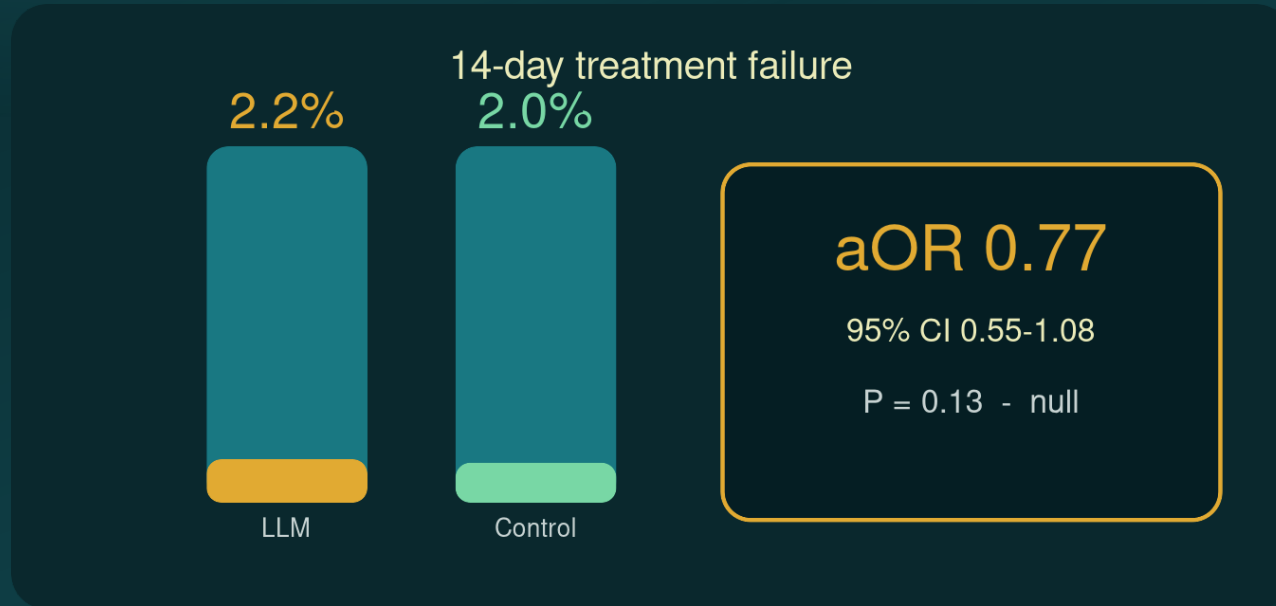


Screened 17,626 → analysed 4,693 vs 4,654. Adults 18–55, 56% women, ~60% febrile/infectious.

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PRIMARY OUTCOME

Treatment failure did not fall



They can reject a revolution. They cannot confirm a nudge.



Table 2 — ITT and per-protocol sit on the same number

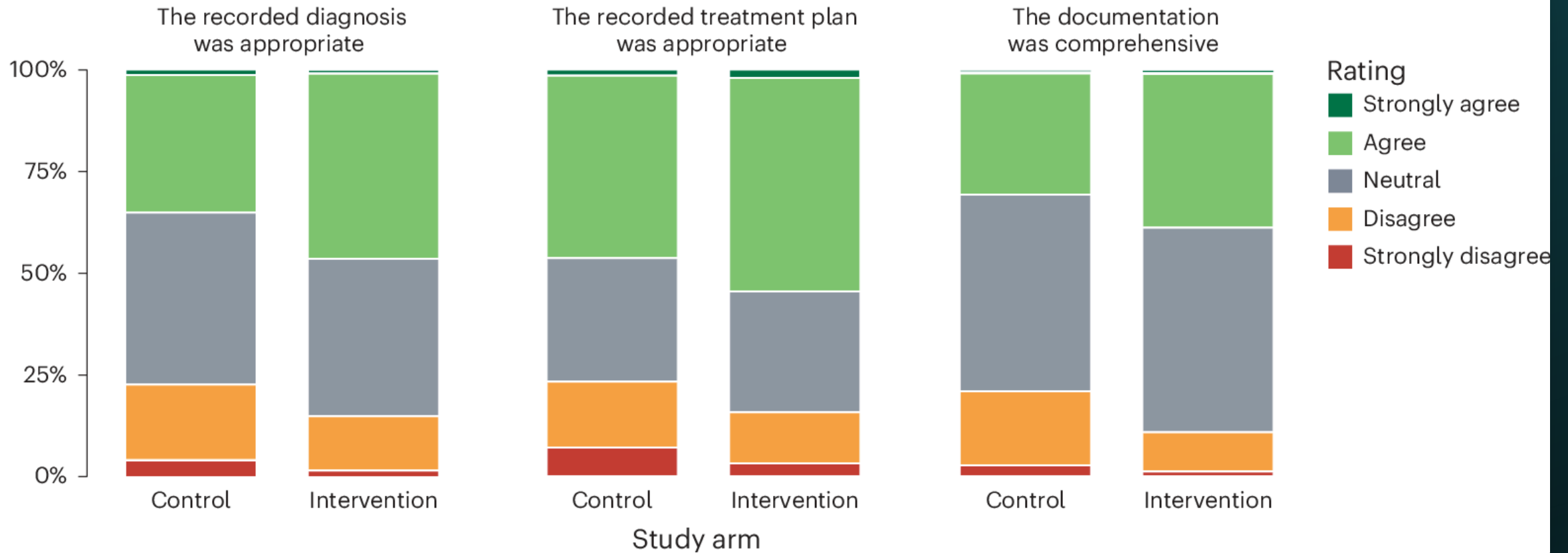
Table 2 | Treatment failure by study arm (intention-to-treat and per-protocol analyses)

Analysis	Characteristic	Failure, <i>n</i> (%)	No failure, <i>n</i> (%)	aOR	95% CI	Pvalue
ITT (adjusted for facility and clinical officer)	Control	94 (2.0)	4,560 (98.0)	0.77	0.55 to 1.08	0.13
	Intervention	102 (2.2)	4,591 (97.8)			
Per protocol (adjusted for facility and clinical officer)	Control	93 (2.0)	4,528 (98.0)	0.77	0.55 to 1.08	0.13
	Intervention	102 (2.2)	4,591 (97.8)			
ITT (cluster and covariate adjusted*)	Control	94 (2.0)	4,560 (98.0)	0.72	0.50 to 1.03	0.07
	Intervention	102 (2.2)	4,591 (97.8)			

Failure counts 94 vs 102. Odds ratio below 1; confidence interval crosses 1.
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Fig. 2a — Notes got better

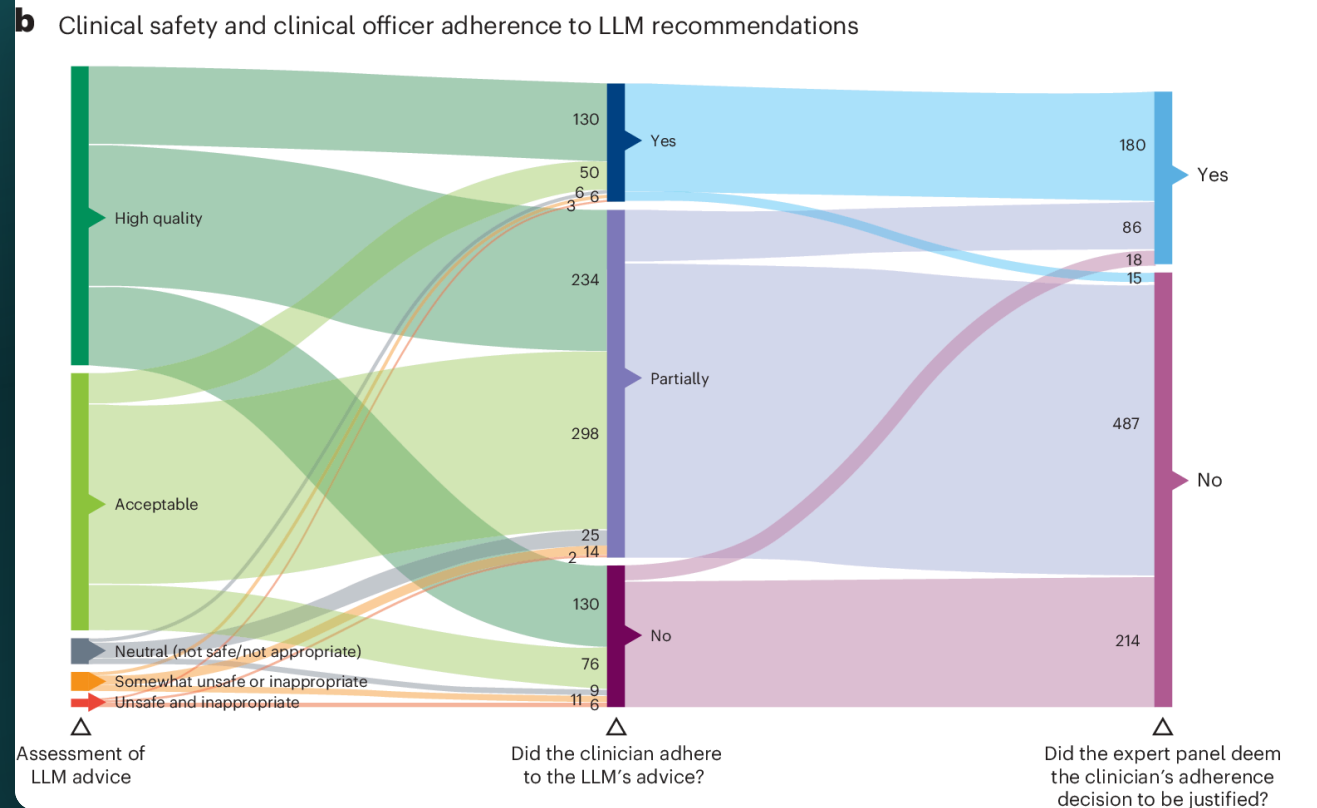
a Quality of the final clinical documentation among control and LLM-assisted consultations by domain



Appropriate diagnosis aOR 1.74 · comprehensive note 1.68 · appropriate plan 1.71 — all $P < 0.001$.

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Fig. 2b — Linger here: a mostly-safe model, a messy team



1,000 red alerts: 91.8% safe-ish · 1.1% unsafe. Followed fully 19.5%. Expert panel: follow-or-ignore **not justified in 71.6%**.

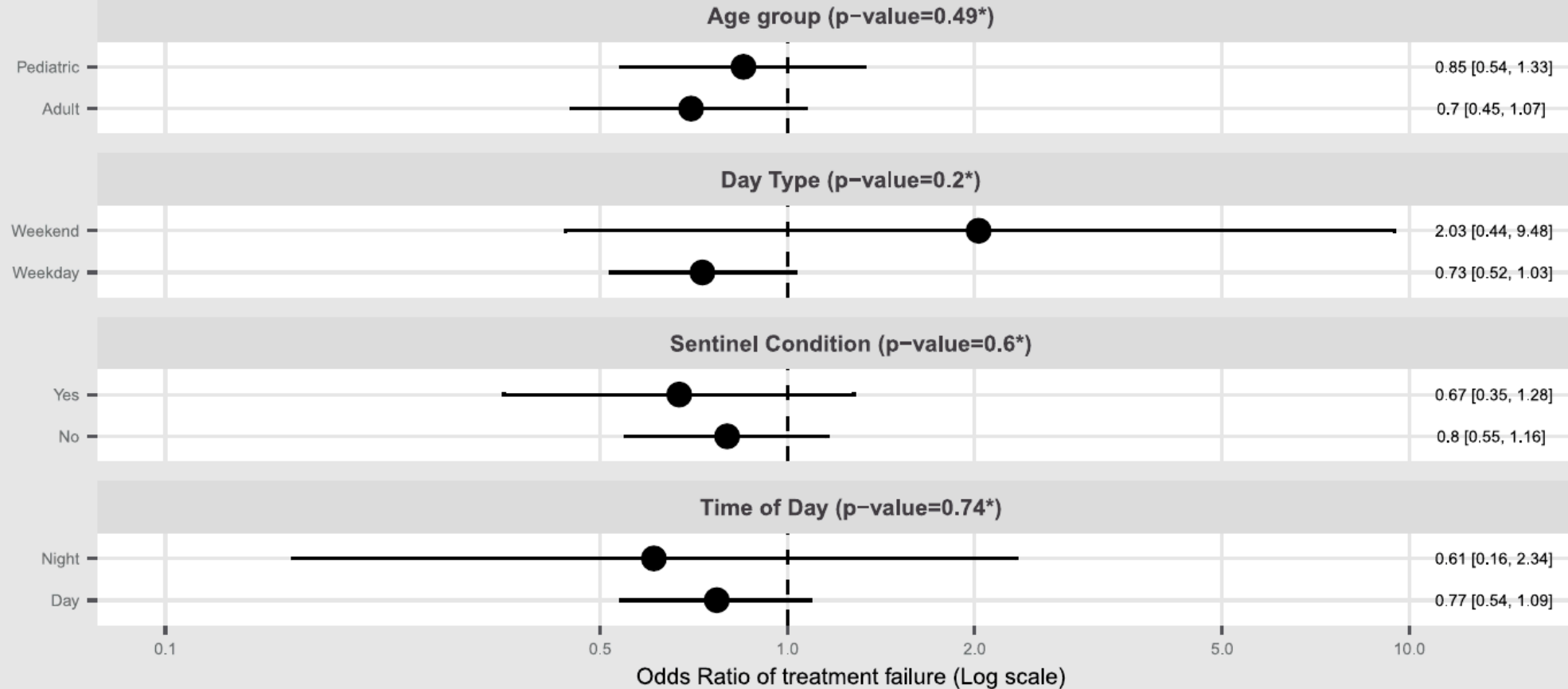
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Fig. 3 — Prescribing and sentinel conditions, mostly null

Prescription: antibiotics Received treatment ≥38.0 °C and prescribed antibiotics: ▶ 135 (of 170; 79%) in control arm ▶ 110 (of 145; 76%) in intervention arm Received proper treatment (including withholding): ▶ 79 (of 170; 46%) in control arm ▶ 61 (of 145; 42%) in intervention arm	Received poor-quality/incorrect treatment Patients for whom antibiotic use was appropriate, but poor-quality/unsafe prescriptions were identified: ▶ 14 (of 48; 29%) in control arm ▶ 5 (of 29; 17%) in intervention arm	Effect of the intervention Correct antibiotic use was similar between groups (OR 0.86; 95% CI 0.48 to 1.55; $P = 0.60$)
Prescription: antimalarials Received proper diagnosis Of patients who received an antimalarial prescription... ▶ 23 (of 29; 79%) in control arm ▶ 26 (of 30; 87%) in intervention arm ...had a malaria test	Received poor-quality/incorrect treatment Patients who received an antimalarial prescription despite a negative test or with no test performed: ▶ 7 (24%) in control arm ▶ 6 (20%) in intervention arm	Effect of the intervention The intervention did not reduce incorrect antimalarial use compared with control (OR 0.76, 95% CI 0.17 to 3.43, $P = 0.70$)
Diagnosis: acute malnutrition Received proper diagnosis Children under 5 years assessed and diagnosed with malnutrition: ▶ 43 (of 120; 36%) in control arm ▶ 49 (of 130; 38%) in intervention arm	Received appropriate referral Eligible children referred to a nutritionist: ▶ 35 (of 120; 29%) in control arm ▶ 44 (of 130; 34%) in intervention arm	Effect of the intervention The odds of receiving a malnutrition diagnosis did not differ between study arms (OR 0.91, 95% CI 0.50 to 1.64, $P = 0.70$) Referral to a nutritionist was also similar across study arms. (OR 1.14, 95% CI 0.61 to 2.13, $P = 0.70$)
Diagnosis: type 2 diabetes mellitus Symptoms identified Patients classified as at risk for diabetes: ▶ 1,578 (of 2,896; 54%) in control arm ▶ 1,447 (of 2,904; 50%) in intervention arm Received proper investigations Identified as at risk for diabetes and tested: ▶ 92 (of 1,578; 5.8%) in control arm ▶ 74 (of 1,447; 5.1%) in intervention arm	Received proper treatment Of those patients diagnosed with diabetes... ▶ 25 (of 33; 76%) in control arm ▶ 24 (of 32; 75%) in intervention arm ...started treatment	Effect of the intervention Those classified as at risk for diabetes were lower in the intervention arm than in the control arm (OR 0.88, 95% CI 0.78 to 0.98, $P = 0.023$) Among those identified as at risk, diabetes testing was infrequent and comparable between groups (OR 0.84, 95% CI 0.56 to 1.26, $P = 0.40$) Treatment initiation among those diagnosed with diabetes was also similar in both groups (OR 0.80, 95% CI 0.01 to 59.3, $P > 0.90$)
Diagnosis: hypertension Received diagnosis Patients who received a new hypertension diagnosis: ▶ 153 (of 2,783; 5.5%) in control arm ▶ 132 (of 2,799; 4.7%) in intervention arm	Received proper treatment Patients with a new hypertension diagnosis who started treatment: ▶ 36 (of 153; 23.5%) in the control arm ▶ 23 (of 132; 17.4%) in the intervention arm	Effect of the intervention Odds of receiving a new hypertension diagnosis were comparable across the control and intervention arms (OR 0.85, 95% CI 0.67 to 1.08, $P = 0.20$) Among patients newly diagnosed with hypertension, treatment initiation was also similar across arms (OR 0.68, 95% CI 0.37 to 1.23, $P = 0.20$)

Antibiotics, antimalarials, HTN, malnutrition: no clear effect. Fewer “at risk of T2D” labels (aOR 0.88, $P = 0.023$).

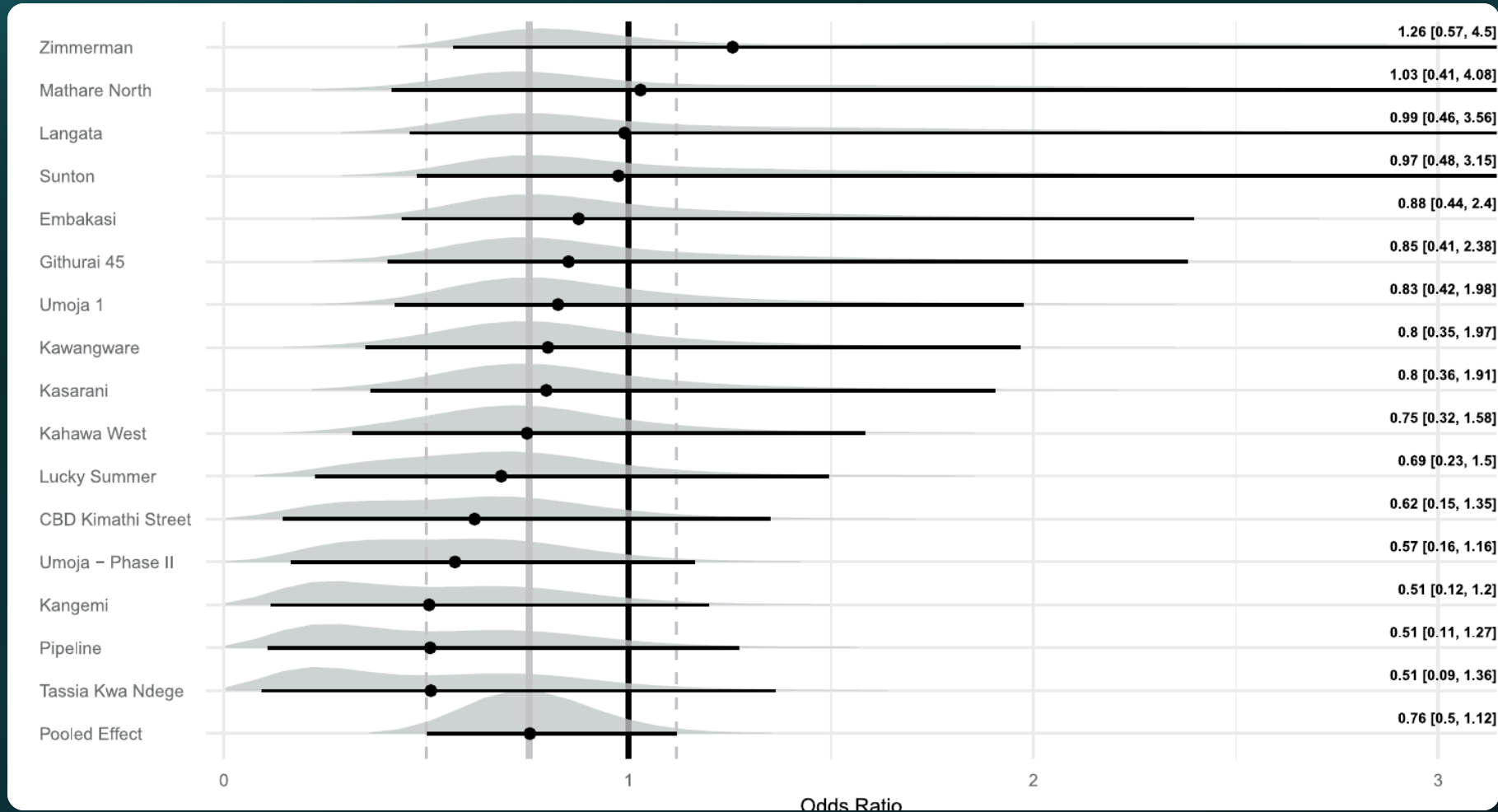
Extended Data Fig. 1 — No subgroup rescues the primary



Age, weekend, sentinel condition, night vs day. Interactions are noisy.

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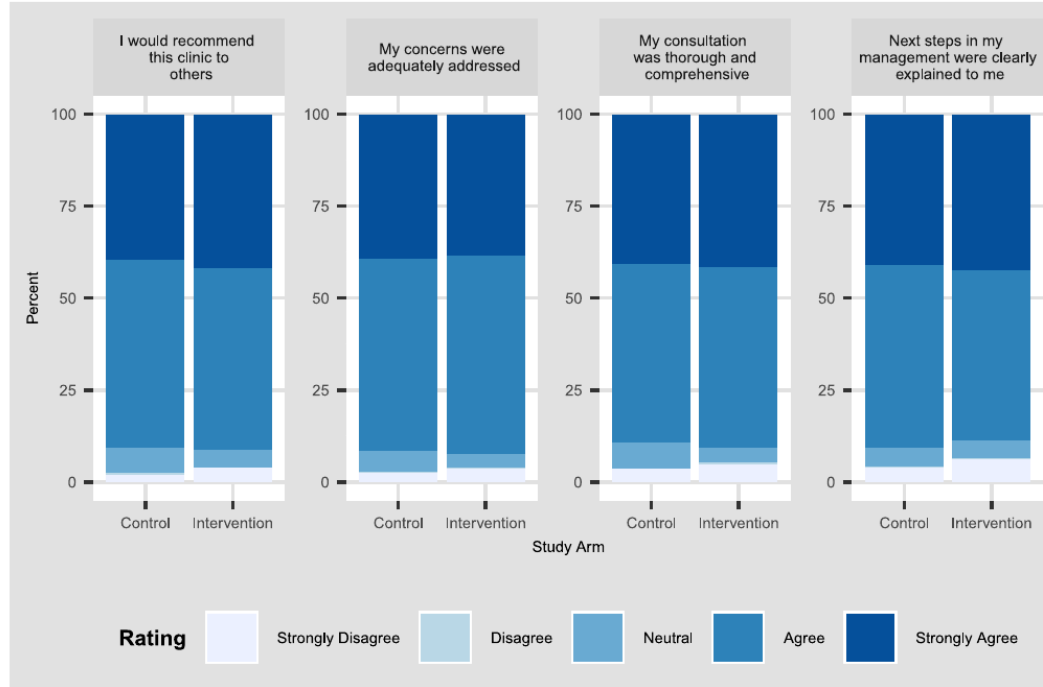
Extended Data Fig. 2 — Same direction, too few events



14 / 16 sites point toward benefit. Pooled OR 0.76 (95% CrI 0.50–1.12).

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Extended Data Fig. 3 — Patients did not notice



Median 4/5 both arms. Visit 11 min. 33 SAEs judged unrelated.
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INTERPRETATION

Why the primary can be null



Powered for 50%

Observed effect, if any, is a nudge.



Multi-cause

Housing, follow-up, the next clinician.



Optional use

Effectiveness, not enforced efficacy.



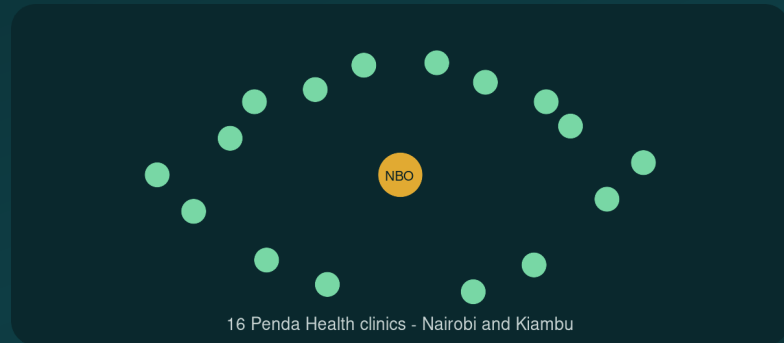
High baseline

Penda already audits and digitizes.



14 days

Note quality might pay off later — or never.





WHAT I AM TAKING INTO NEXT WEEK

Process moved. Outcomes did not — yet

PROCESS

Notes up

diagnosis, plan, completeness

OUTCOMES

Failure flat

14-day failure unchanged

Override **rate** is easy. Override **quality** is Fig. 2b.
Deskilling is unnamed in the data.

YOUR TURN

Paper club questions



1. Next endpoint?

Which primary, and can you actually power it?



2. Better notes

Success, stepping stone, or the wrong target?



3. Override

Healthy skepticism vs a tool too easy to ignore?



4. Where next?

Public-sector and rural replications before scale?

TAKE-HOME

Three lines



Safe*

No related SAE signal at this scale.



Better notes

Diagnosis, plan, completeness.



Null primary

14-day failure unchanged.



Modest?

Large benefit unlikely.

***Not a formal non-inferiority safety analysis.**



EPISODE 1 · NOT MEDICAL ADVICE · A CAREFUL READING

See you next week

Paper Agweyu et al., *Nat Med* 2026 · CC BY 4.0

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You record and speak

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